

Tutorial 7(b)
Kepler's Law of Motion

1. Vector Calculus : Velocity and Acceleration in Polar Coordinates

(a) Given that $\hat{r} = \cos\theta\hat{i} + \sin\theta\hat{j}$, the unit radial vector show that

(i) $\frac{d\hat{r}}{dt} = \dot{\hat{r}} = \frac{d\theta}{dt}(-\sin\theta\hat{i} + \cos\theta\hat{j}) = \dot{\theta}\hat{\theta}$ where $\hat{\theta} = -\sin\theta\hat{i} + \cos\theta\hat{j}$ is the unit tangential vector.

(ii) $\frac{d\hat{\theta}}{dt} = \dot{\hat{\theta}} = -\dot{\theta}\hat{r}$

(iii) $\hat{r}$ and $\hat{\theta}$ are orthonormal vectors.

(iv) $\hat{r}$ and $\dot{\hat{\theta}}$ are orthogonal vectors.

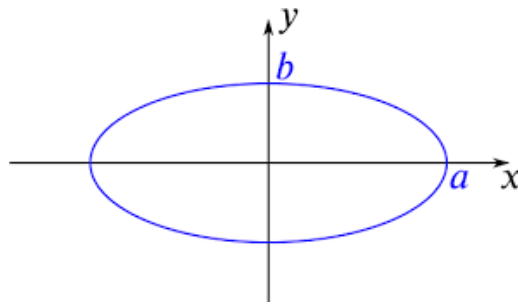
(b) Given also a vector $\vec{r} = r\hat{r}$ in polar coordinates, show that

(i) $\vec{v} = \frac{d\vec{r}}{dt} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta}$

(ii) $\vec{a} = \frac{d\vec{v}}{dt} = (\ddot{r} - r\dot{\theta}^2)\hat{r} + (2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\theta}$ and hence write down the acceleration vector for circular motion.

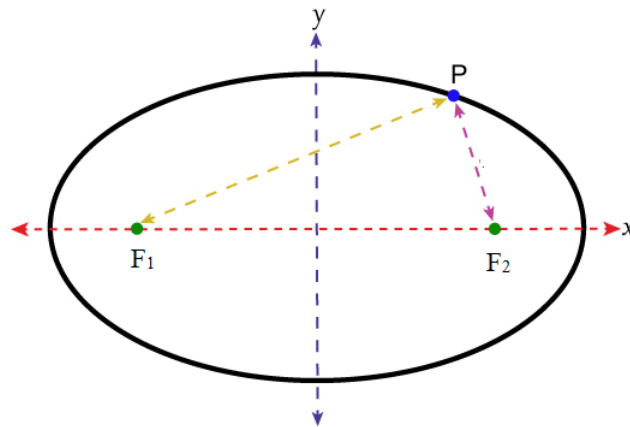
2. Ellipses

- (a) Given that an ellipse centre at the origin has equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ where a is the semi major axis and b is the semi-minor axis.



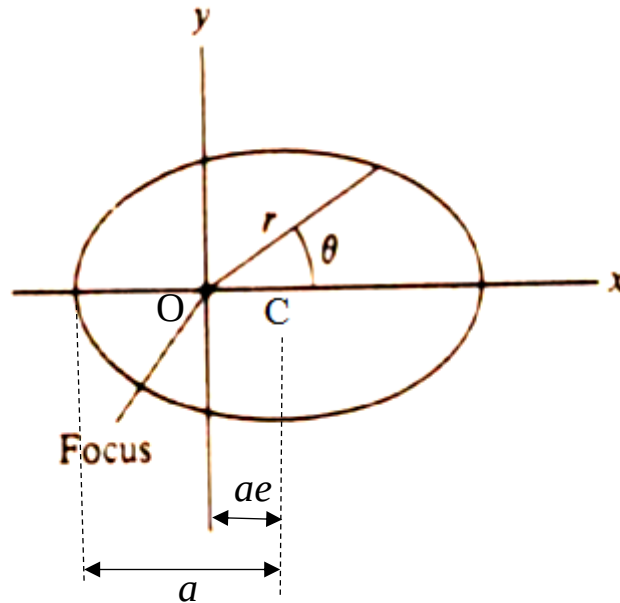
By using calculus or otherwise, show that the area of the ellipse is πab .

- (b) An ellipse has two focal points F_1 and F_2 with the property that $F_1P + F_2P$ is a constant for all point P on the ellipse.



Show that $F_1P + F_2P = 2a$ where a is the semi-major axis.

- (c) Suppose we have an ellipse with its origin O at the left focal point. The point C is the centre of the ellipse, a is the semi-major axis and $0 < e < 1$ known as the eccentricity as shown in the diagram below.



Write down the cartesian equation of the ellipse with its origin O at the left focal point. By using a suitable substitution or otherwise, show that the polar equation of an ellipse with its origin at the left focal point is given by

$$r = \frac{r_0}{1 - e \cos \theta}$$

where $r_0 = a(1 - e^2)$ is known as the semi-latus rectum. By using a simple horizontal translation $\pm ae$ argument, explain why the polar equation of the ellipse with its origin at the right focal point is given by

$$r = \frac{r_0}{1 + e \cos \theta}$$

By using some geometrical reasoning together with the result in (b) show also that $b^2 = a^2(1 - e^2)$ and thus $b^2 = ar_0$ where b is the semi-minor axis.

3. Proof of Kepler's First Law using the Energy method.

(a) Recall a planet subjected to an inverse square law attractive force has energy E given by the equation

$$E = \frac{1}{2}mv^2 - G\frac{Mm}{r} \quad (1)$$

and central force motion gives constant angular momentum

$$L = I\omega = mr^2\dot{\theta} \quad (2)$$

By using the velocity vector $\vec{v}$ obtained in 1(b)(i) and the substitution $\rho = \frac{1}{r}$ show that

$$\dot{r}^2 = \frac{2E}{m} + 2GM\rho - \frac{L^2}{m^2}\rho^2 \quad (3)$$

By completing the square of (3), show that

$$\dot{r}^2 = \left(\frac{L}{m}\right)^2 \left[\left(\frac{e}{r_0}\right)^2 - \left(\rho - \frac{1}{r_0}\right)^2 \right] \quad (4)$$

where $r_0 = \frac{L^2}{GMm^2}$ and $e^2 = 1 + \frac{2Er_0}{GMm}$

To obtain the trajectory of the planetary motion, we consider (2):

$$\frac{d\theta}{dt} = \dot{\theta} = \frac{L}{mr^2} \quad \Rightarrow \quad \theta = \int \frac{L}{mr^2} dt$$

By using chain rule and the given substitution $\rho = \frac{1}{r}$ show that

$$\theta = -\frac{L}{m} \int \frac{d\rho}{\dot{r}} \quad (5)$$

(b) By using the substitution $x = a\cos\theta$ or otherwise show that

$$-\int \frac{dx}{\sqrt{a^2 - x^2}} = \cos^{-1}\left(\frac{x}{a}\right) + c \quad (6)$$

where c is the constant of integration.

(c) Suppose we set up the coordinate axes such that when $\theta = \frac{\pi}{2}$, $r = r_0$, by using result (4), (5) and (6) show that

$$r = \frac{r_0}{1 + e\cos\theta}$$

4. Planetary Motion – Conics Section

We have shown in question 3 that a central force motion subjected to attractive inverse square law has trajectory

$$r = \frac{r_0}{1 + e \cos \theta} \quad (1)$$

where

$$r_0 = \frac{L^2}{GMm^2} \quad \text{and} \quad e^2 = 1 + \frac{2Er_0}{GMm} \quad (2)$$

(a) Show that the polar equation (1) can be written in cartesian form as

$$(1 - e^2)x^2 + 2r_0ex + y^2 = r_0^2$$

(b) From the cartesian equation, determine the type of the planet's trajectory when

- (i) $e = 0$
- (ii) $0 < e < 1$
- (iii) $e = 1$
- (iv) $e > 1$

Note:

- (i) The value e is known as the eccentricity and r_0 semi-latus rectum.
- (ii) Equation (2) from energy method reveal that the planet's trajectory type depends on the total energy of the orbiting planet E , its angular momentum L and the mass of both the central planet M and orbiting planet m .